TOTAL INTEGRABILITY OF A MINKOWSKI BILLIARD IN ITS UNIT BALL

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Abstract. Let a smooth strongly convex norm on the plane define both a billiard table and the length of its chords. We prove that if the entire billiard phase annulus is foliated by continuous essential invariant curves, then the norm comes from an inner product. No finite symmetry beyond central symmetry is assumed. The proof retains the support point of each chord as an independent variable, converts length variations into alternating area variations, and averages over the two resulting phase annuli. A centro-affine boundary parameter and the Blaschke–Santaló inequality then force the unit ball to be an ellipse.

1. Introduction

In a Minkowski billiard, straight chords are measured by a norm and the reflection law is obtained by stationarity of their total norm length. When the table is the unit ball of that same norm, one may ask whether a foliation of all reflected trajectories detects an inner-product norm. This is the rigidity question attributed to Y. Ostrover in [4, Section 9.5, Question 1].

Let $N : \mathbb{R}^2 \to [0, \infty)$ be a norm, smooth away from zero, with $D^2(N^2)$ positive definite there, and put $K = \{N \le 1\}$. Let $\gamma(s)$ parametrize ∂K by positive N-arclength, with period T. The billiard map acts on ordered distinct boundary points, represented by $s < t < s + T$, and has generating function

$$L(s, t) = N(\gamma(t) - \gamma(s)).$$

Its consecutive parameters satisfy

$$L_2(s_{j-1}, s_j) + L_1(s_j, s_{j+1}) = 0.$$

We use *total integrability* to mean that this entire phase annulus is foliated by continuous invariant simple closed curves winding once around it.

Bialy and Tsodikovich proved rigidity under an additional finite linear symmetry of order at least three [5, Theorem 5]; their formulation uses a C^1 foliation. Their argument uses periodic invariant curves to produce progressively more symmetries. A general norm ball has central symmetry, of order two, which does not supply that initial higher-order symmetry. The following theorem proves the rigidity implication for continuous invariant foliations without the additional assumption.

Theorem 1.1. *For every norm N as above, if its billiard in $K = \{N \le 1\}$ is totally integrable, then K is an origin-centered ellipse. Equivalently, there exists a real symmetric positive-definite matrix Q such that*

$$N(v) = \sqrt{v^T Q v} \qquad \text{for every } v \in \mathbb{R}^2.$$

The proof uses an area variational description in position and momentum. This description and its relation to symplectic billiards were studied by Albers, Chavez Caliz, and Tabachnikov [1, Sections 2–3]. For a single symplectic billiard table, total-integrability rigidity was proved by Baracco and Bernardi [2, Theorem 1.1]. Here the position curve and the rotated polar momentum curve need not coincide, so a single-table rigidity theorem does not directly

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apply. Instead, each norm chord is the maximum of an area expression over its support point. This maximum transfers finite variational inequalities to an alternating sequence on the two curves. Averaging requires both alternating phase annuli.

The remaining estimate is geometric. We choose a boundary parameter for which the signed area integral of the derivative curve equals that of the original curve. Its period is at most 2π by the Blaschke–Santaló inequality. The averaged variational inequality and an energy lower bound then force equality in that inequality. The assumption concerns the whole phase annulus; integrability only near its boundary is outside the theorem. Section 2 develops the alternating variational inequality, and Section 3 proves the geometric estimates and the theorem.

2. From norm length to alternating area

Throughout, set

$$\omega(x,y) = x_1y_2 - x_2y_1, \qquad J(x_1,x_2) = (-x_2,x_1).$$

For a convex body containing the origin, its polar is $C^* = \{p : p \cdot x \le 1 \text{ for every } x \in C\}$. Let

$$P = -JK^*, \qquad A = \operatorname{area}(K), \qquad B = \operatorname{area}(P) = \operatorname{area}(K^*). \tag{1}$$

Both boundaries are smooth, have positive curvature, and are centrally symmetric. All boundary parameters increase counterclockwise.

2.1. Support points and finite variations.

Keeping the support point of a chord as a separate variable turns its norm length into a sum of two signed areas.

Lemma 2.1. *For distinct $x, z \in \partial K$,*

$$N(z-x) = \max_{y\in\partial P}\{\omega(x,y) + \omega(y,z)\}. \tag{2}$$

The maximizer is unique and has strictly negative second derivative in any regular boundary parameter of P.

Proof. The expression being maximized is $(z-x)\cdot Jy$, and $JP = K^*$. The support function of K^* is N. For instance, convexity and homogeneity show that $DN(z-x) \in K^*$ and $DN(z-x)\cdot(z-x) = N(z-x)$. Positive curvature of the polar gives uniqueness and nondegeneracy: at the supporting point a nonzero outward supporting functional has strictly negative second derivative along the boundary. □

We first record the variational consequence of total integrability. For a positive regular parameter γ on ∂K, put $w = \gamma(t) - \gamma(s)$. Convexity and homogeneity give $D^2N(w)w = 0$. Moreover $D^2N(w)$ is positive on every direction transverse to w. Otherwise, subtracting a multiple of w would give a nonzero u with $DN(w)u = 0$ and $D^2(N^2)(w)[u,u] = 0$. Thus, in the plane,

$$D^2N(w)[u,v] = \lambda(w)\omega(w,u)\omega(w,v), \qquad \lambda(w) > 0.$$

At the two endpoints of a chord, $\omega(\gamma'(s),w) > 0$ and $\omega(\gamma'(t),w) < 0$. Consequently the norm-length generating function has

$$L_{12}(s,t) = -D^2N(w)[\gamma'(s),\gamma'(t)] > 0. \tag{3}$$

In the original N-arclength parameter, $-L_1(s,t)$ decreases from 1 to -1 as t runs from s to $s+T$. These coordinates identify the ordered-point annulus with the usual symplectic twist cylinder.

For this sign convention, every orbit on a continuous essential invariant curve is locally maximizing for the action with fixed endpoints; see [3, Section 3, paragraph after Theorem 3.1, p. 5]. The hypothesis of Theorem 1.1 therefore makes the Hessian of every finite length action negative semidefinite. This property is unchanged by a smooth increasing boundary reparametrization: at a stationary action its Hessian changes by diagonal congruence.

For even vertices $q_i \in \partial K$ and odd vertices $p_i \in \partial P$, consider

$$\sum_i \{\omega(q_i, p_i) + \omega(p_i, q_{i+1})\}. \tag{4}$$

At the supporting p_i, maximizing each odd variable in (4) gives exactly the original length action. The odd-variable Hessian is diagonal negative definite by Lemma 2.1. If the full finite Hessian is $\begin{pmatrix} U & V \\ V^T & W \end{pmatrix}$, its Schur complement $U - VW^{-1}V^T$ is the Hessian of that length action. Completing the square,

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix}^T \begin{pmatrix} U & V \\ V^T & W \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \xi^T (U - VW^{-1}V^T)\xi + (\eta + W^{-1}V^T\xi)^T W(\eta + W^{-1}V^T\xi) \le 0.$$

Every finitely supported alternating variation can be enclosed between fixed even endpoints. Hence all its second variations are nonpositive.

2.2. **The two half maps.** Let $c(s)$ and $d(t)$ be positive regular parametrizations of ∂K and ∂P. Define the two annuli

$$\mathcal{D}_{KP} = \{(s,t) : \omega(c'(s), d'(t)) > 0\}, \qquad \mathcal{D}_{PK} = \{(t,s) : \omega(d'(t), c'(s)) > 0\}.$$

For $(q,p) = (c(s), d(t)) \in \mathcal{D}_{KP}$, let z be the other intersection of ∂K with the ray $q + ud'(t)$, $u > 0$, and set $S(q,p) = (p,z)$. The determinant condition means the ray enters K. It exits transversely at z, where $\omega(d'(t), c'(z)) > 0$; here $c'(z)$ denotes the positive tangent at z. Thus the image belongs to $\mathcal{D}_{PK}$. On this component define $S(p,z) = (z, p_+)$ using the other intersection of P with the ray from p in the positive tangent direction at z. Reverse chords give the inverse half maps. Transversality makes them smooth. We regard S as one invertible map of the disjoint union $\mathcal{D} = \mathcal{D}_{KP} \sqcup \mathcal{D}_{PK}$.

The point p is the maximizer for $N(z - q)$ in (2). Indeed the outward normal to P at p is a positive multiple of $-Jd'(t)$, and $z - q$ is a positive multiple of $d'(t)$. Conversely every ordered distinct q, z determines exactly such a p. At the next even vertex, the stationarity equation

$$\omega(c'(z), p_+ - p) = 0$$

is, by differentiation of (2), the norm-length reflection equation. Its outgoing branch is the other intersection $p_+ \neq p$: the equal intersection would continue the incoming chord out of K. Therefore S^2 on $\mathcal{D}_{KP}$ is conjugate to the original Minkowski billiard. Every edge in either component lies on an alternating lift of an original orbit. The finite second-variation conclusion above holds on the whole union.

On each component use $H(u,v) = \omega(u,v)$, differentiating in its respective boundary parameters, and write $b = H_{12} > 0$. The measure

$$d\mu = b\, ds\, dt \tag{5}$$

on each component is finite and the combined measure on $\mathcal{D}$ is invariant under S. To see this directly, differentiate $H_2(s,t) + H_1(t,s_+) = 0$ and wedge with dt:

$$H_{12}(s,t)\, ds \wedge dt = H_{12}(t,s_+)\, dt \wedge ds_+.$$

Both densities are positive on their corresponding annuli. Finiteness follows from bounded derivatives on the compact parameter tori.

2.3. **An averaged inequality.** At n consecutive interior vertices choose variation 1, with variation 0 at the two external endpoints. The nonpositive second variation is

$$\sum_{j=1}^{n} \{H_{22}(u_{j-1}, u_j) + H_{11}(u_j, u_{j+1})\} + 2\sum_{j=1}^{n-1} H_{12}(u_j, u_{j+1}) \le 0.$$

Integrate over the initial edge in the invariant union $\mathcal{D}$. There are n diagonal terms and $n-1$ adjacent pairs, so invariance gives

$$n\int_{\mathcal{D}}(H_{11}+H_{22})\,d\mu+2(n-1)\int_{\mathcal{D}}H_{12}\,d\mu\le 0.$$

All integrands are bounded and the measure is finite. Dividing by n and letting $n\to\infty$ yields

$$\int_{\mathcal{D}}(H_{11}+2H_{12}+H_{22})H_{12}\,ds\,dt\le 0. \tag{6}$$

Choose the parameters to respect antipodes: $c(s+\ell_c/2)=-c(s)$ and $d(t+\ell_d/2)=-d(t)$. Such parameters will be constructed in Lemma 3.1. Replacing one endpoint by its antipode reverses every second derivative of H. It leaves the quadratic integrand in (6) unchanged and exchanges the positive- and negative-determinant domains. The determinant-zero integrand vanishes. Thus each component integral is half of its full product-torus integral. Interchanging the two bodies negates H, interchanges H_{11},H_{22}, and leaves that product unchanged. The two full integrals agree. Consequently (6) is precisely

$$\iint\{\omega(c'',d)+2\omega(c',d')+\omega(c,d'')\}\omega(c',d')\,ds\,dt\le 0, \tag{7}$$

with both parameters running through full periods.

We express this in terms of the energy matrices

$$\begin{aligned} M_c&=\int c'c'^{T}\,ds, \qquad M_d=\int d'd'^{T}\,dt,\\ C_c&=\int\omega(c',c'')\,ds, \qquad C_d=\int\omega(d',d'')\,dt,\\ D&=\iint\omega(c',d')^2\,ds\,dt=\operatorname{tr}(M_cJM_dJ^T). \end{aligned} \tag{8}$$

Recall that $A=\operatorname{area}(K)$ and $B=\operatorname{area}(P)=\operatorname{area}(K^*)$. Periodic integration by parts and the area formula give $\int dd'^{T}\,dt=-BJ$ and $\int cc'^{T}\,ds=-AJ$. At fixed s the first term in (7) therefore integrates to $-B\omega(c',c'')$; interchanging the bodies gives the other term. It follows that

$$2D\le BC_c+AC_d. \tag{9}$$

3. Affine estimates and the equality case

To use (9), we choose parameters for which each derivative integral C_c,C_d is twice the area of its original body. The same parameters have a universal period bound.

Lemma 3.1. *Let $C=-C$ be a smooth planar convex body with positive curvature and area $a>0$. There is a positive boundary parameter σ, of period $\Omega_C\le 2\pi$, such that*

$$\int_0^{\Omega_C}\omega(q_\sigma,q_{\sigma\sigma})\,d\sigma=2a. \tag{10}$$

Antipodal points differ by $\Omega_C/2$ in this parameter.

Proof. First use the centro-affine area parameter r, defined by $\omega(q,q_r)=1$. It exists because the origin is strictly interior; its full period is $2a$. Differentiation and positive curvature give

$$q_{rr}=-k(r)q, \qquad k(r)>0.$$

The polar boundary is $v=-Jq_r$. Indeed v is the outward supporting covector at q and $v\cdot q=1$. Since $v_r=kJq$, $\omega(v,v_r)=k$, so, with $a^*=\operatorname{area}(C^*)$,

$$\int k(r)\,dr=2a^*.$$

Set

$$d\sigma = \sqrt{k(r)}\,dr, \qquad \Omega_C = \int \sqrt{k(r)}\,dr.$$

Cauchy–Schwarz and the symmetric Blaschke–Santalo inequality [6, Theorem 1] give

$$\Omega_C^2 \le 4aa^* \le 4\pi^2. \tag{11}$$

Direct differentiation gives

$$q_\sigma = \frac{q_r}{\sqrt{k}}, \qquad q_{\sigma\sigma} = -q - \frac{k_r}{2k^2}q_r, \qquad \omega(q_\sigma, q_{\sigma\sigma}) = \frac{1}{\sqrt{k}}.$$

Integrating proves (10). Central symmetry gives $q(r+a) = -q(r)$ and $k(r+a) = k(r)$, so antipodes are half a σ period apart. □

Lemma 3.2. *Let q be a smooth regular simple closed planar curve, enclosing area $a > 0$, in a parameter of period ℓ. Then its energy matrix $M = \int q'q'^T$ satisfies*

$$\sqrt{\det M} \ge \frac{2\pi a}{\ell}. \tag{12}$$

Proof. The matrix M is positive definite: a vector in its kernel would make one component of q constant and put the curve in a line. The area-preserving linear map $T = (\det M)^{1/4}M^{-1/2}$ satisfies $TMT^T = \sqrt{\det M}\,I$. If l is the Euclidean perimeter of Tq, Cauchy–Schwarz and ordinary isoperimetry [7, Section 1, equation (1.5)] imply

$$4\pi a \le l^2 \le \ell \int |Tq'|^2 = 2\ell\sqrt{\det M}.$$

□

Proof of Theorem 1.1. Parametrize both boundaries by Lemma 3.1. In (8) this gives $C_c = 2A$, $C_d = 2B$, with periods $\Omega_K, \Omega_P \le 2\pi$. The variational inequality (9) becomes

$$D \le 2AB. \tag{13}$$

On the other hand, for positive-definite 2 by 2 matrices U, V,

$$\operatorname{tr}(UV) \ge 2\sqrt{\det U \det V},$$

by the arithmetic–geometric mean inequality for the eigenvalues of $U^{1/2}VU^{1/2}$. Lemma 3.2 applied separately to c, d therefore gives

$$D = \operatorname{tr}(M_c J M_d J^T) \ge 2\sqrt{\det M_c \det M_d} \ge \frac{8\pi^2 AB}{\Omega_K \Omega_P} \ge 2AB. \tag{14}$$

The separate linear maps used to prove (12) establish scalar determinant bounds; the mixed trace in (14) is computed in the original coordinates.

Since $A, B > 0$, (13)–(14) force $\Omega_K\Omega_P = 4\pi^2$ and hence $\Omega_K = \Omega_P = 2\pi$. Since $B = \text{area}(K^*)$, applying (11) to $C = K$ yields $4\pi^2 \le 4AB \le 4\pi^2$. Equality in the symmetric Blaschke–Santalo inequality makes K an origin-centered ellipse. Its gauge is $N(v) = \sqrt{v^TQv}$ for a real symmetric positive-definite matrix Q, including $v = 0$. □

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